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# A bayesian ordinal structural equation modeling framework for evaluating the seismic resilience of IoT-enabled buildings under uncertainty

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ABSTRACT

Internet of Things (IoT) enabled earthquake resilience has advanced through real time sensing, Earthquake Early Warning (EEW), and adaptive structural control using devices such as active mass dampers and semi active damping systems. Yet, decision grade integration with code relevant performance assessment under uncertainty remains limited. This study develops a unified, uncertainty-consistent framework for evaluating and improving the earthquake resilience of IoT-enabled buildings in terms of code relevant performance. Prior work is often fragmented, frequently treats ordinal resilience and compliance indicators as continuous, and commonly applies simplified treatments of sensing and warning uncertainty that weaken performance-based decision making. A Bayesian Ordinal Structural Equation Modeling (BOSEM) framework is proposed that jointly integrates real time IoT-based structural assessment with survey and inspection-based latent variable resilience modeling, linking observed streaming signals and ordered categorical indicators within a single probabilistic system. The approach combines an ordinal measurement layer based on threshold likelihoods for survey and inspection indicators with observed streaming inputs from IoT and early warning systems, including noise, missingness, latency, packet loss, lead time, predicted intensity, uncertainty bounds, and contextual building attributes. Outputs include a code relevant resilience outcome and decision relevant posterior quantities, enabling end to end uncertainty propagation from sensing and warning through latent mechanisms to compliance relevant outcomes. The framework is demonstrated on 162 cases. Measurement results indicate strong ordinal quality, with composite reliabilities ranging from 0.83 to 0.92 and average variance extracted ranging from 0.51 to 0.65. Discriminant validity is supported by Heterotrait Monotrait ratio of correlations (HTMT) posterior means from 0.53 to 0.63, with upper credible bounds not exceeding 0.72. Posterior predictive adequacy is supported by Posterior Predictive Check (PPC) tail area probability 0.46, polychoric residual Root Mean Square Error (RMSE) 0.07, maximum absolute item level PPC residual 0.12, and share of indicators with residual exceeding 0.10 equal to 0.14. Structural inference identifies RTAQ as the strongest direct driver of Seismic Resilience Performance (SRP) with an effect of 0.38, followed by Adaptive Control Effectiveness (ACE) 0.31, with EWR 0.21 and Code Structural Integrity (CSI) 0.19, and Pr (beta $>$ 0) from 0.971 to 0.999. Model comparison favors the full specification, including response style and cross loadings, achieving Pareto Smoothed Importance Sampling Leave One Out Information Criterion (PSIS LOOIC) 4121.6, Widely Applicable Information Criterion (WAIC) 4124.0, and a stacking weight of 0.75, while endogeneity diagnostics indicate near zero residual correlations and instrumental relevance, with conditional R squared values of 0.16 for Real Time Assessment Quality (RTAQ) and 0.12 for CSI.

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## 1. Introduction

The IoT market in the construction sector is projected to reach approximately USD 16 billion by 2024, corresponding with a Compound Annual Growth Rate (CAGR) of 26.0%. This rapid expansion is driving widespread adoption of IoT-enabled technologies, particularly in applications related to structural health monitoring and seismic resilience, where real-time sensing and data-driven analysis support improved infrastructure safety and performance. A prominent example is the expanding deployment of sensors in Structural Health Monitoring (SHM) systems. The global SHM market is expected to grow at a CAGR of 17.9% between 2017 and 2024 [1]. In parallel, IoT enabled EEW systems, such as ShakeAlert, developed by the U.S. Geological Survey, aim to mitigate earthquake impacts by providing warning of seconds to minutes before strong ground shaking. Retrofitting initiatives incorporating IoT technologies have demonstrated measurable improvements in building resilience in earthquake prone regions. IoT applications are also projected to reduce construction costs by 20%-25%, supporting long-term cost-effectiveness for seismic strengthening strategies [2]. Collectively, these developments underscore the growing recognition of IoT as a key enabler of resilient and sustainable infrastructure worldwide.

Adaptive structural control represents the operational mechanism through which IoT based sensing and EEW information can be transformed into tangible reductions in seismic demand. In smart base isolated buildings, semi-active or adaptive devices such as Magneto Rheological (MR) dampers can be tuned in real time to balance isolation displacement and superstructure acceleration, thereby improving structural performance under varying earthquake intensities [3]. Recent studies indicate that advanced control strategies can outperform conventional passive systems, motivating integrated consideration of warning reliability and control effectiveness within seismic resilience frameworks [4]. Despite these advances, significant research gaps remain in integrating IoT technologies into seismic resilience. Although civil engineering practice has increasingly shifted toward strategies that enhance earthquake resistance in vulnerable buildings, and resilience has received heightened attention in active seismic zones, the effects of IoT based sensing and warning systems on building performance have not been thoroughly validated through empirical and probabilistic analyses. A central challenge is the treatment of uncertainty associated with hazard occurrence and system response [5].

Prior research on disaster preparedness has often used risk assessment and forecasting approaches based on historical earthquake data [6] While valuable, such approaches can be limited in their ability to represent uncertainty explicitly. Subsequent studies have introduced Bayesian methods for uncertainty analysis, including Bayesian modal identification with fast optimization validated through simulation and experiments, demonstrating improved accuracy and robustness under uncertainty. Bayesian networks have also been applied to uncertainty prediction in disaster related contexts, further supporting probabilistic inference in seismic analysis [7]. In parallel, prior research has emphasized the need to redefine seismic design objectives. A resilience-oriented design perspective argues that criteria should extend beyond life safety to include post-earthquake usability and operational continuity [8]. This emphasis reflects observations that many buildings do not collapse in major earthquakes yet still require demolition rather than repair, creating substantial economic and social losses [9]. The resilience oriented perspective has also been linked to sustainability goals, reinforcing the need for performance oriented evaluation metrics [10].

From a regulatory perspective, seismic resilience should be framed in terms of explicit performance objectives. Performance Based Earthquake Engineering (PBEE) formalizes these objectives through measurable limits, such as interstory drift ratio, floor acceleration, and functional performance states across hazard levels [11]. Accordingly, IoT and EEW enabled resilience strategies should be evaluated by their ability to increase the probability of meeting code and standards based performance targets [12]. At the same time, earthquake prediction and uncertainty modeling studies have explored advanced Bayesian networks and community-based forecasting strategies, reporting accuracy improvements of up to 65% compared with traditional approaches [2]. However, these efforts typically focus on prediction performance and do not explicitly link uncertainty aware inference to real time building response, adaptive control actions, or regulatory performance outcomes. Moreover, key constructs in resilience and compliance, such as preparedness, warning credibility, compliance maturity, and operational readiness, are often measured using ordered categorical indicators. Treating ordinal responses as continuous can bias estimation and distort relationships, motivating ordinal latent variable formulations for statistically valid inference. The need to bridge these gaps is reinforced by the operational constraints of EEW systems, which typically provide only seconds to tens of seconds of lead time before strong ground shaking [13]. Under such conditions, resilience enhancing actions require rapid, uncertainty aware inference. Recent IoT edge cloud architectures for earthquake monitoring emphasize real time analytics that translate streaming sensor data into actionable decisions within strict timing windows. Practical deployments also introduce latency, synchronization error, missing data, and communication unreliability, motivating explicit modeling of data acquisition and transmission quality in resilience frameworks [6,14].

Researchers in 2018 reported that earthquake forecasting accuracy can be improved by dividing a region into communities, generating forecasts within each community, and combining results to form a final prediction [15]. A Bayesian network has also been incorporated into an earthquake network prediction framework using a dataset spanning roughly 20 years of earthquake occurrences, with a reported prediction success rate of about 65%, higher than baseline approaches in that study [16]. Beyond civil infrastructure, IoT has been used for real time monitoring and traceability in aquaculture systems, illustrating sensing, transmission, and user facing access to field data. IoT data streams have also been applied to wind energy prediction tasks, illustrating real time sensing combined with predictive analytics [17]. Soil Structure Interaction (SSI) can materially alter structural demand and perceived performance by altering the effective period, damping, and foundation compliance of the coupled soil foundation structure system. Recent experimental and numerical evidence indicates that foundation and connection details within superstructure pile foundation systems can significantly modify response quantities used in performance assessments, suggesting that demand and compliance predictions can be biased when the soil foundation interface is idealized [18]. Interaction driven mechanisms, such as foundation level separation and pounding sensitivity, further indicate that coupled response effects can change drift, and acceleration demands in ways not captured by fixed base assumptions.

In parallel, passive resilience strategies remain a dominant stream for mitigation and assessment. These include deterioration aware resilience assessment frameworks that quantify how ageing and degradation shift resilience metrics over the service life. Complementary work on recovery function-based resilience quantification emphasizes that resilience is governed not only by peak response but also by post event recovery trajectories and functionality restoration. Passive geotechnical mitigation measures similarly target permanent deformation and serviceability loss through ground improvement rather than real time actuation. The broader resilience literature also frames these issues in multi-hazard settings, where indirect losses and functional disruptions arise from interacting hazards and recovery constraints [19]. The present study is distinguished from both streams by a decision grade formulation that integrates real time sensing and early warning information with an ordinal latent variable representation of readiness mechanisms and a probabilistic mapping to code relevant compliance outcomes under uncertainty. While SSI and passive systems primarily influence demand through physical coupling, degradation, or fixed mitigation, the

proposed BOSEM framework focuses on end-to-end uncertainty propagation from information quality to latent mechanisms and then to posterior compliance quantities that support operational decision making within real time feasibility limits.

Existing research addresses key components of earthquake resilient buildings but does so in fragmented and methodologically limited ways. EEW studies translate warning information into expected building response and potential actions, and adaptive base isolation studies use intensity estimates to tune devices such as MR dampers. Yet uncertainty is often handled via simplified deterministic sensitivity assumptions rather than probabilistic propagation into decisions and outcomes. In parallel, IoT driven seismic resilience studies using Structural Equation Modeling (SEM) identify important constructs but typically rely on non-Bayesian SEM and often treat Likert type indicators as continuous, limiting validity for ordinal measurements and constraining uncertainty quantification. Consequently, there is no unified framework that simultaneously models ordinal resilience indicators correctly, propagates uncertainty from IoT sensing and EEW forecasts into adaptive control actions, and links these actions to code relevant performance outcomes. This study addresses this gap by developing a Bayesian ordinal SEM that integrates IoT sensing, EEW reliability, and adaptive control within a single probabilistic model to produce posterior decision grade estimates of resilience mechanisms and performance and compliance outcomes under uncertainty.

Building on these gaps, recent advances in physically interpretable machine learning, uncertainty aware identification, and high frequency sensing analytics provide a technically consistent pathway for embedding IoT/EEW information into decision grade seismic resilience models. Physically interpretable feature learning such as wavelet guided networks with dynamic frequency decomposition demonstrates how vibration driven signals can be mapped to robust diagnostics while preserving mechanistic meaning, which is directly relevant to extracting actionable indicators from SHM streams under time constraints [6]. At the regional to asset scale, machine learning coupled with Geographic Information System (GIS) based seismic vulnerability modeling illustrates how heterogeneous exposure, site effects, and inventory characteristics can be fused into predictive vulnerability layers that can serve as informative priors or covariates within Performance-Based Earthquake Engineering (PBEE) oriented inference [20]. For structure level response reconstruction, adaptive measurement based substructure identification supports fast dynamic state estimation from partial observations, enabling rapid updating of demand estimates immediately after EEW triggers [21,22], while interval deep neural networks with explicit damage functions provide a principled route to "credible" identification by bounding epistemic uncertainty an essential requirement when warning reliability and communication loss are non-negligible [10,23]. Bayesian data fusion approaches in geotechnical characterization further show how uncertain soil parameters (e.g., fabric/void ratio proxies) can be probabilistically consolidated, aligning with the need to represent SSI sensitive inputs as posterior distributions rather than fixed values [24,25]. Complementarily, multi sensor fault diagnosis frameworks using variational mode decomposition with multi scale convolutional models and graph based sensor selection demonstrate how IoT networks can be optimized for reliability and latency by prioritizing informative channels and filtering nonstationary noise, which directly informs modeling of data quality and missingness in real deployments [26]. For mapping high-dimensional indicators (including ordinal readiness constructs) to performance outcomes, hierarchical fuzzy topological regression offers an interpretable nonlinear surrogate that can be embedded as a component model or benchmark for latent mechanism inference [27]. On the actuation side, fast neural dynamic solvers developed for real time control problems indicate how control relevant computations can be executed within stringent timing windows, supporting uncertainty aware adaptive device tuning immediately following EEW updates [28]. Robust decision making under coupled uncertainties is further motivated by stochastic robust optimization frameworks (e.g., bi level hybrid game formulations) that formalize adversarial variability and operational constraints concepts transferable to resilience decisions that must trade off competing objectives such as drift, acceleration, and downtime under uncertain hazards and information quality [29]. Finally, component level evidence from seismic analyses of composite systems with concrete filled square steel tubular columns provides the structural performance grounding needed to ensure that inferred compliance states and recovery relevant metrics reflect realistic nonlinear behavior of contemporary structural typologies [30]. Collectively, these works justify an integrated Bayesian ordinal SEM in which (i) interpretable signal decomposition and multi sensor optimization inform the measurement model, (ii) Machine Learning (ML)/GIS vulnerability and substructure/interval identification inform the probabilistic demand model under epistemic and aleatory uncertainty, (iii) Bayesian fusion supports SSI consistent parameter inference [24], (iv) real time control and robust optimization inform the decision layer for adaptive isolation actions, and (v) experimentally/analytically supported structural typologies anchor the mapping from posterior demands to code relevant compliance outcomes.

To further operationalize the proposed BOSEM perspective within code relevant decision making, the probabilistic compliance mapping must be anchored to structural typologies and component behaviors that are routinely assessed using nonlinear response history methods and validated connection/section mechanics. Incremental Dynamic Analysis (IDA) of lateral systems, including buckling restrained braced frames and special moment resisting frames, provides a direct route to estimating demand to capacity exceedance probabilities across intensity measures, thereby supporting posterior inference of performance state attainment under uncertainty and enabling consistent comparison of alternative control and retrofit actions within PBEE style objectives [31,32]. At the component and member scale, numerical and experimental evidence on welded built up Concrete Filled Tubular (CFT) columns covering axial compressive behavior sensitivity to welding line configuration, fabrication details, and cold-formed section assembly clarifies how strength, ductility, and stability limits should be parameterized in the structural model so that posterior compliance predictions do not rely on idealized section behavior [33,34]. System-level resilience and operational continuity can be governed by connection performance and deformation compatibility; accordingly, investigations of through-beam connections to Concrete-Filled Circular Steel Tube (CFCST) columns, together with complementary finite element studies of through-beam connections in CFSST–steel beam assemblies, establish mechanics-based constraints on joint stiffness, local failure modes, and hysteretic degradation that are required to be reflected when inferred demands are translated into functional performance outcomes [35,36]. In the same spirit, validated finite element procedures (e.g., ABAQUS based beam modeling) provide a defensible pathway for representing Reinforced Concrete (RC) member nonlinearities and damage evolution, which is essential when coupling IoT/EEW updated demand estimates to component level limit states in a Bayesian setting [37]. For braced systems whose global response depends on brace activation asymmetry and load path intermittency, seismic performance factor evaluation for tension only braced frames further informs how overstrength, ductility, and redundancy parameters should be treated probabilistically rather than deterministically when deriving code related quantities from posterior response predictions [38]. Extending beyond conventional building frames, computational studies of specialized structural forms (e.g., cylindrical barns) highlight that geometric nonlinearity and shell like dynamics can materially alter demand distributions, reinforcing the need for model class aware uncertainty propagation when BOSEM is applied to diverse infrastructure assets [39]. Finally, evidence on supplemental damping devices such as pre bent strip dampers provides an additional passive/semiactive benchmark for how energy dissipation mechanisms modify drift and acceleration demands, thereby offering a comparative baseline against which IoT/EEW triggered adaptive control

benefits can be quantified in posterior compliance and functionality metrics [40].

To strengthen the decision-grade grounding of the proposed BOSEM framework, the structural performance layer should be calibrated against empirically supported failure mechanisms and member–system interactions observed across modern structural and infrastructure typologies. Progressive collapse sensitivity in multistory RC frames highlights how localized member loss can trigger disproportionate system response, motivating probabilistic coupling between latent readiness mechanisms and robustness-oriented limit states in the compliance mapping [41]. For steel and composite framing, experimentally validated cyclic behavior in steel–PEC spliced beam systems and composite braced frames provides directly transferable parameters for hysteretic degradation and residual demand, ensuring that posterior performance states remain compatible with component-level evidence [42,43]. Bridge and long-span dynamics studies on arch–beam joints and vortex-induced vibration suppression using advanced inertial damping devices further justify embedding controllable dissipation mechanisms (passive-to-active) as comparable baselines when quantifying incremental benefits of EEW-triggered adaptive actions [44,45]. Because BOSEM also targets operational continuity under uncertainty, multiscale collapse predictions for RC sub-assemblages can inform epistemic bounds on damage evolution and repairability, especially when sensor data quality is imperfect or delayed [46]. Real-world geotechnical–structural coupling is also essential: deformation monitoring of tunnels under excavation and the associated uncertainty in ground–structure interaction provide a practical analogue for modeling observation noise, missingness, and site-driven bias in SHM streams [47]. For distributed lifeline assets, self-sensing electronic gaskets for pipeline SHM and uncertainty-aware quantitative risk models offer complementary evidence for representing IoT-enabled condition states and ordinal readiness indicators as latent variables linked to decision-grade risk outputs [48,49].

In underground construction, overviewed jacking-force calculation methods and data-driven settlement prediction for rectangular pipe-jacking tunnels reinforce the need to treat soil–structure interaction inputs as posterior distributions rather than fixed values, particularly when real-time decisions rely on partial observations [50,51]. Reliability of IoT sensing networks depends on the robustness of field materials and device environments; nonlinear dynamic strength criteria for concrete that explicitly account for ultimate dynamic strength support incorporating material-level resistance characteristics into PBEE consistent priors for demand capacity inference [52]. At the solver level, return-mapping algorithms with hyper-dual derivative approximation enable stable elastoplastic updates under rapid computation, which is consistent with BOSEM's requirement for latency-feasible inference following EEW triggers [53]. Functional failure mechanisms observed in rectangular underground structures during earthquakes further provide structure-class-specific constraints on serviceability and functionality states used in resilience-oriented compliance evaluation [54]. For sensing and predictive assessment in tunnels, machine learning finite element mesh optimization based modeling and prediction of excavation-induced shield tunnel ground settlement, together with coupled FDM–DEM excavation simulations, illustrate how high frequency data streams and physics informed computational models can be transformed into actionable indicators while explicitly acknowledging modeling and measurement uncertainty [55,56].

In operational deployment, shield parameter prediction via multivariate variable-weight combination models offers a concrete template for integrating heterogeneous covariates into real-time updating when site and construction conditions evolve [57]. Multi-hazard considerations are supported by studies of residential natural-gas explosion damage patterns, which motivate extending BOSEM's ordinal readiness–compliance link to compound disruption pathways that affect recovery and usability beyond peak response [58]. For optimization under uncertainty, quantum mapping approaches to non-probabilistic reliability and seismic design studies of self-centering prestressed concrete frames with sliding infill walls under earthquake loading support treating control decisions as constrained updates driven by uncertain hazard and state information [59,60]. Ground motion feature uncertainty can be reduced using lagged coherence estimation, enabling more credible intensity-to-demand mappings when EEW inputs are noisy or partially synchronized [61]. For interface-sensitive seismic systems, wave propagation analyses of layered media with imperfect interfaces emphasize that interface condition and incident wave characteristics can significantly shift inferred seismic response, validating BOSEM's explicit propagation of SSI-related epistemic uncertainty into compliance probabilities [62]. System-level dissipation evidence from precast segmental CFST bridge piers with external arched links provides additional benchmarks for comparing passive/semiactive energy pathways against IoT/EEW-driven adaptive strategies [63]. In distributed sensing, Markov Chain Monte Carlo (MCMC)-based relocation and time correction for ocean-bottom seismographs directly supports modeling synchronization error and timing uncertainty, which are critical for early-warning credibility and rapid posterior updating [64]. Privacy-preserving federated stochastic primal–dual learning for Internet-of-Vehicles illustrates a deployable pathway for secure multi-node learning, aligning with BOSEM's need to fuse distributed IoT streams without compromising data governance [65]. For aerodynamic–structural interaction under forced vibration, large-eddy simulations of lock-in dynamics demonstrate that nonlinear excitation mechanisms can shift response distributions, reinforcing the necessity of robust uncertainty modeling when translating sensor-informed predictions to code-relevant states [66]. Finally, attribute- and instance-weighted Naive Bayes provides a lightweight probabilistic classification baseline that can be used to benchmark BOSEM's latent ordinal inference when rapid, interpretable decision rules are required under tight EEW timing windows [67].

## 2. Methodology

### 2.1. Study framework and objectives

The study aims to (1) construct a Bayesian ordinal measurement and structural model for key latent resilience mechanisms data acquisition & transmission quality, early warning reliability, real time analytics capability, ACE/readiness, and code/standards integration consistent with established SEM best practice guidelines for latent variable specification and identification; (2) quantify and propagate uncertainty from sensing and EEW/ Seismic Early Warning System (SEWS) predictions to control decisions in accordance with Bayesian inference standards, including explicit prior specification, posterior uncertainty quantification, and probabilistic sensitivity analysis; and (3) estimate posterior probabilities of meeting code relevant performance targets following the principles of PBEE, where compliance is evaluated relative to measurable demand thresholds rather than qualitative safety criteria [17].

The primary inputs (exogenous observed variables) include IoT sensing metrics (noise/precision indicators, missing data rate, latency/jitter, packet loss, calibration/health status), EEW/SEWS metrics (predicted intensity and arrival time, lead time, uncertainty bounds/credible intervals, alert type), and contextual covariates (building typology/height/period class, isolation system characteristics, occupancy/importance class), defined to explicitly capture communication and sensing constraints relevant to real time inference and control. Key outputs (endogenous variables) include control decision variables (e.g., control mode selection such as passive, semi active, or adaptive passive; MR damper command level; isolation tuning state) and performance/compliance outcomes, represented as continuous risk metrics (e.g., probability of exceeding drift or acceleration limits) and/or ordinal performance states aligned with PBEE performance objectives (e.g., low, moderate, or high impact corresponding to accepted performance levels). These objectives are operationalized through a staged

methodology encompassing system boundary definition, variable specification, Bayesian ordinal measurement modeling, structural uncertainty propagation, and probabilistic performance evaluation, as detailed in Sections 2.2–2.7, ensuring consistency with the proposed Bayesian ordinal modeling workflow and PBEE-oriented performance evaluation framework. nformed decision support for smart a.

### 2.2. Study system, boundary conditions, and experimental design

The study system is defined as the closed decision and response chain that begins with EEW issuance and ends with post-action evaluation of building response against code relevant performance criteria. The system boundary includes data acquisition from distributed sensing, edge level preprocessing and transmission, early warning feature ingestion, inference of latent resilience mechanisms, and selection of control actions that affect the response. Exogenous inputs comprise time indexed sensing and communication quality descriptors, as well as early warning descriptors. In contrast, endogenous outputs comprise control decisions and the resilience outcome, represented as an ordinal code aligned with a state or an exceedance probability for demand thresholds [68].

#### 2.2.1. System boundary and time discretization

The analysis window is defined on $t \in [0, T]$ from early warning issuance to the end of response evaluation, and it is discretized at a resolution $\Delta t$ to match sensing cycles and edge processing cycles. Time is indexed as

$$t_k = k\Delta t, k = 0, 1, \ldots, K, T = K\Delta t \tag{1}$$

The boundary conditions enforce feasibility under communication and sensing constraints at each discrete time. Latency and packet loss are bounded as

$$0 \le L_k \le L_{\max}, 0 \le p_{\text{loss},k} \le p_{\max} \tag{2}$$

where $L_k$ denotes the end to end latency from measurement to availability for inference and $p_{\text{loss},k}$ denotes the packet loss probability at time $t_k$. These bounds define the admissible operating region for reliable inference and for control actions that require timely updates [69].

#### 2.2.2. Control and operational constraints

Physical device limits and operational safety requirements constrain control actions. At each time step, the control input is restricted to

$$u_k \in [u_{\min}, u_{\max}] \tag{3}$$

where $u_k$ denotes the command applied to the adaptive device, and the bounds reflect hardware capacity and stability requirements. This constraint ensures that inferred decisions are physically implementable. Operational feasibility is further constrained by the available early warning lead time. Control actions must be computed and executed before strong motion arrival, after accounting for communication delay, expressed as

$$\tau_k^{\text{act}} \le \tau_k^{\text{lead}} - L_k \tag{4}$$

where $\tau_k^{\text{act}}$ is the required time for inference and actuation, and $\tau_k^{\text{lead}}$ is the warning lead time. This condition couples sensing quality and communication performance to control viability and explicitly embeds real time limitations within the model structure [5].

### 2.3. Variables, inputs, and outputs, and latent construct definition

#### 2.3.1. Input vectors and preprocessing

All observed quantities entering the model are organized as a time indexed feature vector representing the information available for inference and decision making at time $t_k$. The input vector is defined as

$$x_k = [s_k, w_k, c] \tag{5}$$

where $s_k$ denotes IoT sensing and communication descriptors, $w_k$ denotes early warning descriptors, and $c$ denotes contextual attributes that remain constant over the analysis window. The sensing component $s_k$ includes indicators of measurement noise, data missingness, timing jitter, packet loss probability, and sensor health status. These variables characterize sensing and communication conditions at each time step and directly influence the information available for inference. The early warning component includes predicted ground motion intensity, waveform based source characterization descriptors, estimated arrival time, available lead time, and uncertainty related warning information [70]. The contextual component $c$ includes building typology, height, and period class, isolation system properties, and occupancy or importance class. Preprocessing is limited to scaling, time alignment, and aggregation choices needed to ensure that all elements of $x_k$ are defined on the discretized grid $t_k$, while preserving the ordinal or probabilistic nature of the original measurements.

#### 2.3.2. Latent resilience constructs and output variables

Latent resilience mechanisms are represented by a vector of unobserved constructs defined as

$$\eta = [\eta_{RTAQ}, \eta_{EWR}, \eta_{ACE}, \eta_{CSI}, \eta_{SRP}]^\top \tag{6}$$

where $\eta_{RTAQ}$ denotes real time analytics quality, $\eta_{EWR}$ denotes early warning reliability, $\eta_{ACE}$ denotes adaptive capability, $\eta_{CSI}$ denotes sensing and communication integration, and $\eta_{SRP}$ denotes the latent resilience outcome construct [71]. This construct set is consistent with the measurement and structural results reported in Tables 1–3, where RTAQ, EWR, ACE, CSI, and SRP are all treated as latent variables supported by ordinal indicators, with discriminant validity assessed through HTMT, and with structural effects estimated toward SRP. Model outputs consist of operational decisions and performance or compliance outcomes available at each time step, expressed as

$$y_k = [d_k, u_k, r_k, z_k] \tag{7}$$

where $d_k$ is the discrete control mode selection, $u_k$ is the applied control command, $r_k$ is a continuous risk-based metric, such as a demand exceedance probability, and $z_k$ is an ordinal performance state aligned with code-relevant performance objectives. This output structure enables linkage between latent mechanisms $\eta$, operational decisions, and probabilistic performance evaluation, while maintaining consistency between the latent resilience outcome construct $\eta_{SRP}$ used in the structural system and the decision and compliance quantities defined at the output layer.

#### 2.3.3. Sensor attachment integrity and long-term durability

Field performance of IoT-based structural monitoring depends not only on sensor specifications but also on the integrity of the sensor to substrate attachment. In buildings, attachment degradation can arise from ageing, environmental exposure, cyclic vibration, surface cracking, adhesive creep, and local spalling, which may introduce bias and

**Table 1**
Bayesian posterior reliability and convergent validity (N = 162).

| Latent construct | Avg. posterior loading | Composite reliability (95% CrI) | AVE (95% CrI) |
|---|---|---|---|
| RTAQ | 0.75 | 0.88 [0.84, 0.92] | 0.58 [0.50, 0.65] |
| EWR | 0.70 | 0.83 [0.78, 0.88] | 0.51 [0.43, 0.59] |
| ACE | 0.77 | 0.89 [0.85, 0.93] | 0.60 [0.52, 0.67] |
| CSI | 0.72 | 0.86 [0.81, 0.90] | 0.54 [0.46, 0.62] |
| SRP | 0.80 | 0.92 [0.88, 0.95] | 0.65 [0.57, 0.72] |

**Table 2**
Posterior HTMT ratios (upper triangle, mean with 95% CrI).

| | RTAQ | EWR | ACE | CSI |
|---|---|---|---|---|
| RTAQ | | 0.59 [0.49, 0.69] | 0.63 [0.54, 0.72] | 0.56 [0.46, 0.66] |
| EWR | | | 0.57 [0.47, 0.67] | 0.53 [0.43, 0.63] |
| ACE | | | | 0.60 [0.50, 0.70] |
| CSI | | | | |

**Table 3**
Standardized Bayesian structural effects on SRP.

| Predictor → SRP | Posterior mean | 95% CrI | Posterior SD | Pr ($\beta > 0$) |
|---|---|---|---|---|
| RTAQ | 0.38 | [0.23, 0.53] | 0.08 | 0.998 |
| EWR | 0.21 | [0.07, 0.36] | 0.07 | 0.982 |
| ACE | 0.31 | [0.18, 0.45] | 0.07 | 0.999 |
| CSI | 0.19 | [0.05, 0.33] | 0.07 | 0.971 |

nonstationary noise in measured acceleration, strain, or displacement proxies [72]. These mechanisms can produce apparent changes in dynamic features that are not attributable to structural response, including amplitude attenuation, phase distortion, intermittent dropouts, and elevated measurement variance. Consequently, attachment durability is a first order determinant of data credibility in long duration deployments. It should be treated as an explicit contributor to sensing quality rather than an implicit assumption.

In the proposed framework, attachment integrity is incorporated through sensor health descriptors within the sensing and communication input vector, for example, diagnostics reflecting bond condition, calibration factor stability, and time varying signal to noise indicators. Deterioration in attachment quality, therefore, enters the model as degraded sensing quality and increased uncertainty, which widens posterior dispersion in the sensing and communication integration construct and propagates to compliance probability through the latent structural system and performance mapping [73]. In practice, this treatment ensures that decisions are not driven by potentially spurious signal variations arising from attachment deterioration, and it provides a coherent mechanism for conservative decision making when sensor integrity cannot be verified. In future implementations, data-driven evaluation of bond performance, including machine-learning-based prediction of bond strength and degradation in concrete related attachments, can be used to improve the fidelity of these sensor health descriptors and to support maintenance scheduling for resilient monitoring systems.

### 2.4. Bayesian ordinal measurement model for survey and inspection indicators

#### 2.4.1. Ordinal indicator likelihood using ordered probit or logit

Each survey or inspection item is treated as an ordered categorical realization generated by an underlying continuous response . $y^{(m)} \in \{1, \ldots, J_m\}$ denotes indicator $m$ measuring construct $\eta_{j(m)}$. The observation rule is defined by thresholding the latent response $y^{*(m)}$ as follows.

$$y^{(m)} = r \Leftrightarrow \tau_{m,r-1} < y^{*(m)} \leq \tau_{m,r} \tag{8}$$

where $\{\tau_{m,r}\}_{r=0}^{J_m}$ are ordered cut points satisfying

$$\tau_{m,0} = -\infty < \tau_{m,1} < \cdots < \tau_{m,J_m} = +\infty. \tag{9a}$$

The latent response is linked to the construct via a single factor measurement equation.

$$y^{*(m)} = \lambda_m \eta_{j(m)} + \epsilon_m \tag{9b}$$

For an ordered probit specification, $\epsilon_m \sim \mathcal{N}(0,1)$ fixes the scale of the latent response. For an ordered logit specification, $\epsilon_m$ follows a logistic distribution with unit scale, yielding the analogous likelihood. This formulation preserves the ordinal nature of the indicators and provides a coherent likelihood for measurement uncertainty.

#### 2.4.2. Identification and reliability and validity constraints

Model identification is enforced through a standard scaling constraint within each construct. For each construct $j$, one reference indicator $m_0(j)$ has its loading fixed as

$$\lambda_{m_0(j)} = 1. \tag{10}$$

Latent constructs are assigned centered priors to define location and permit covariance learning,

$$\eta \sim \mathcal{N}(0, \Sigma_\eta). \tag{11}$$

Together, the fixed loading constraint and the centered construct prior anchor the measurement scale, ensure estimability of remaining loadings and thresholds, and allow posterior assessment of measurement quality through the implied reliability and convergent validity of each construct . Fig. 1 presents a resilience assessment framework that integrates IoT sensing data, early warning indicators, and contextual attributes as inputs. These inputs are scaled and aligned to estimate latent resilience constructs, such as real time analytics quality, early warning reliability, adaptive capability, sensing integration, and overall resilience outcome.

### 2.5. Structural model and uncertainty propagation

#### 2.5.1. Structural equation among constructs

The BOSEM core represents directed dependence among latent resilience mechanisms and links observed inputs to these mechanisms through a latent structural system. The construct vector $\eta$ is defined in accordance with the reported construct set $\eta_{RTAQ}, \eta_{EWR}, \eta_{ACE}, \eta_{CSI}, \eta_{SRP}$. A window level summary of the time indexed inputs is denoted by $x$, supporting stable inference over $t \in [0, T]$. The structural model is specified as

$$\eta = B\eta + \Gamma\overline{\chi} + \zeta, \zeta \sim \mathcal{N}(0, \Psi). \tag{12}$$

The matrix B encodes directed construct effects and defines the internal pathway structure, while $\Gamma$ maps aggregated sensing, early warning, and contextual features into the constructs. The disturbance $\zeta$ captures unobserved influences and permits correlated residual structure through $\Psi$. The aggregation $\bar{x}$ is defined to retain decision relevant information and to control dimensionality, and may include summaries such as mean latency, missingness rate, packet loss level, and representative early warning intensity measures.

#### 2.5.2. Early warning intensity uncertainty and sensing and communication influence

Early warning intensity is treated as a noisy estimate of an underlying latent intensity that drives building demand. The relation is modeled as

$$I^{\text{true}} = I^{\text{EEW}} + \delta, \delta \sim \mathcal{N}\left(0, \sigma_I^2\right) \tag{13}$$

This specification introduces an explicit probabilistic layer for early warning uncertainty, with $\sigma_I^2$ representing the intensity error variance. Sensing and communication conditions enter the structural system through the observed input summaries in $\bar{x}$ and their mapping by $\Gamma$. Within the construct set reported in Tables 1–3, sensing and communication integration is represented by $\eta_{CSI}$, and the structural associations of the constructs with the resilience outcome are represented through effects on $\eta_{SRP}$ (Table 3). Consequently, uncertainty in $I^{\text{EEW}}$ and degradation in sensing and communication jointly propagate through the posterior by widening uncertainty in $I^{\text{true}}$ via Eq. (13) and by shifting the posterior of $\eta$ via Eq. (12), which in turn shifts the posterior distribution

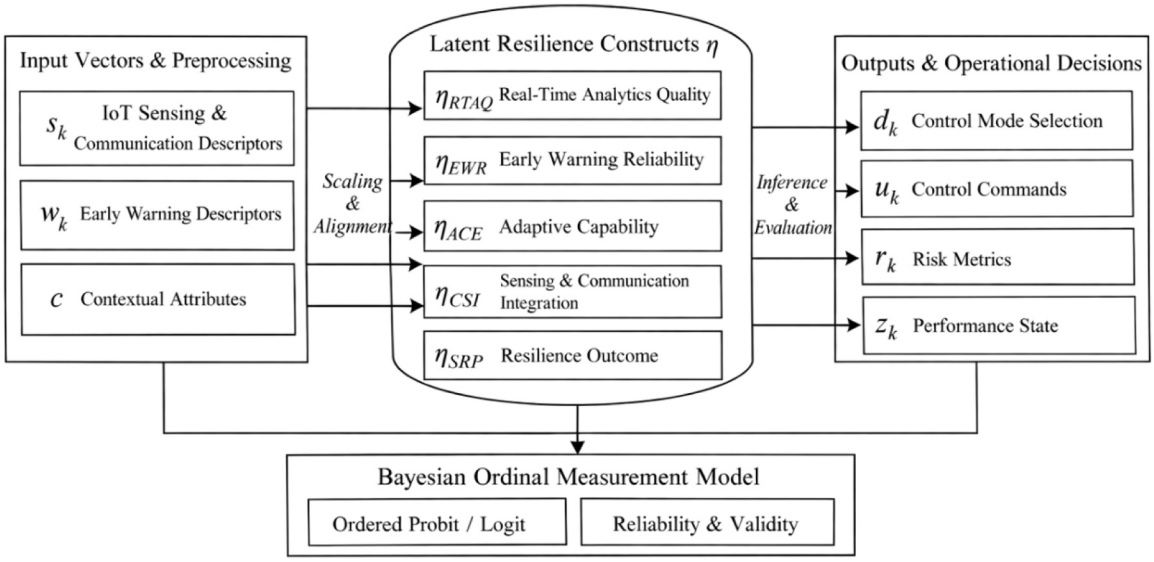



**Fig. 1.** Bayesian ordinal resilience modeling framework integrating IoT sensing inputs, latent resilience constructs, and operational decision outputs.

of decision and performance quantities defined downstream [74].

### 2.6. Decision model, control outputs, and performance and compliance mapping

#### 2.6.1. Probabilistic control mode selection

Control mode selection is modeled as a probabilistic decision layer conditioned on latent resilience mechanisms and current early warning features. At each time step, the control mode $d_k \in \{1,\ldots,D\}$ is selected according to a multinomial logit specification,

$$\Pr(d_k = d|\eta, w_k) = \frac{\exp\left(\alpha_d + \beta_d^\top [\eta, w_k]\right)}{\sum_{d'=1}^{D} \exp\left(\alpha_{d'} + \beta_{d'}^\top [\eta, w_k]\right)} \tag{14}$$

The parameters $\alpha_d$ represent baseline preferences for each control mode, and $\beta_d$ quantify the influence of constructs and warning descriptors. The set of modes includes passive, semi active, and adaptive passive strategies [75]. Construct-level uncertainty is embedded directly into operational decisions, with variability in warning information and sensing or communication conditions introduced, which modify the posterior distribution over selected control actions.

#### 2.6.2. Code relevant performance outcome and exceedance probability

Performance and compliance are evaluated by comparing demand exceedance to a code specified limit. A structural demand measure, such as interstory drift ratio or floor acceleration, is denoted by $D$, and the corresponding limit state threshold is denoted by $D_{\text{lim}}$ . The risk metric is defined as

$$r = \Pr(D > D_{\text{lim}} |\eta, I^{\text{true}}, c) \tag{15}$$

This probability integrates uncertainty in latent resilience mechanisms $\eta$, true seismic intensity $I^{\text{true}}$, and building context $c$. The scalar $r$ represents a decision grade compliance indicator consistent with performance-based evaluation. In addition, the continuous risk measure may be mapped to an ordinal performance state $z \in \{\text{Low}, \text{Moderate}, \text{High}\}$ through predefined probability thresholds aligned with performance objectives. This mapping preserves compatibility with ordinal reporting frameworks while maintaining full probabilistic inference at the latent and demand levels .

### 2.7. Bayesian inference, implementation tools, and validation protocol

#### 2.7.1. Posterior inference and computation

Inference targets the joint posterior over parameters and latent variables implied by the measurement model, the structural model, the intensity uncertainty model, and the decision and outcome mapping. The estimands are collected in $\Theta$.

$$\Theta = \{\lambda, \tau, B, \Gamma, \Sigma_\eta, \Psi, \alpha, \beta, \sigma_I\}. \tag{16}$$

The posterior distribution $p(\Theta, \eta|\mathscr{D})$is obtained using Markov chain Monte Carlo, with a Hamiltonian scheme such as the No U-Turn Sampler, or via variational inference when computational latency constraints dominate . Convergence and sampling adequacy are evaluated using the effective sample size and $\widehat{R}$ statistics, and calibration is assessed through PPCs targeting the ordinal indicator distribution implied by Eqs. (8) and (9) and the compliance and performance outputs implied by Eqs. (14) and (15). This protocol enforces internal consistency between the ordinal measurement layer and the decision grade outcome layer under the same posterior.

#### 2.7.2. Validation protocol, robustness analysis, and boundary condition stress testing

Validation is performed by imposing controlled scenarios that stress the uncertainty and feasibility limits embedded in the system definition. Scenario factors include early warning intensity uncertainty through $\sigma_I$, communication quality through $L_k$ and $p_{\text{loss},k}$, and sensing degradation through missingness and health status embedded in $s_k$. All scenarios are constrained to satisfy the admissible bounds in Eqs. (2) to (4) to preserve the feasibility of inference and actuation [76]. Robustness is evaluated through posterior comparisons of the compliance probability $r$ across control modes $d_k$ and through the sensitivity of structural pathways encoded in $B$. The primary robustness targets are stability of posterior mass for $\Pr(D > D_{\text{lim}}|\eta, I^{\text{true}}, c)$ under variation of uncertainty sources and invariance of dominant construct effects when boundary conditions approach limiting regimes. This testing isolates whether gains in compliance arise from information quality and warning reliability, from adaptive control readiness, or from code and standards integration within the structural system [77].

#### 2.7.3. Practical limitations in IoT deployments: sensor noise, latency, and deterministic baseline validation

*2.7.3.1. Sensor noise, missing data, and communication uncertainty handling.* Real world IoT based monitoring is affected by sensor noise, intermittent missingness, timing jitter, packet loss, and end to end latency. If these effects are handled implicitly, both inference quality and the feasibility of real time control can be overstated. In the proposed framework, these issues are incorporated as explicit data quality descriptors that enter the model through the input vector and the system boundary constraints. Measurement noise is represented through noise sensitive sensing descriptors and the early warning intensity error formulation, so that increased noise or degraded signal quality produces wider posterior uncertainty in the latent mechanisms and in the downstream compliance probability, in line with intelligent diagnostic frameworks developed for noisy sensor signals [78]. At the asset-management level, complementary digital inspection workflows for underground utilities can support continuous updating, permitting, and compliance tracking in networked built-environment systems [79].

Latency and timing jitter is treated in two coupled ways. First, inference quality is degraded through reduced timeliness and coherence of information entering the latent system, as reflected in the communication quality descriptors. Second, operational limits on actuation are imposed through a feasibility condition requiring inference and command execution to occur within available warning lead time. When end to end latency approaches the lead time margin, adaptive actions may remain statistically favorable yet become operationally infeasible. Consequently, a distinction is enforced between posterior preference for a control mode and feasibility of execution, with decisions maintained as consistent with real time cyber physical constraints.

*2.7.3.2. Deterministic baseline validation against Bayesian inference.* A deterministic baseline is defined that uses the same aggregated sensing and early warning summaries. For distributed deployment, secure and verifiable identity-aware data exchange can support traceable handling of these baseline inputs across monitoring nodes [80]. The baseline can be implemented as a deterministic structural response mapping, or a point prediction model calibrated on the same dataset, with traceable data handling preserved for networked IoT deployment settings. Validation is performed by comparing deterministic point estimates of the compliance related output with Bayesian posterior summaries under identical scenarios, including nominal conditions and stressed conditions with elevated noise and increased latency. Agreement is assessed by verifying that deterministic predictions fall within Bayesian credible intervals at the expected frequency and by comparing scenario induced trends across the two approaches [81]. Discrepancies are interpreted with respect to the identified practical limitations: divergence that increases systematically with noise, missingness, or latency is treated as evidence that the Bayesian model is appropriately widening uncertainty and reducing decision certainty under degraded information quality, in line with hybrid monitoring approaches designed for mixed binary and continuous process variables [82]. This baseline comparison provides an external check on plausibility while preserving the primary advantage of the Bayesian formulation, namely, end to end uncertainty quantification from sensing and warning through latent mechanisms to code relevant compliance outcomes [83].

#### 2.7.4. Artificial intelligence (AI) optimization for improved real time inference and control readiness

The proposed framework is agnostic to the specific learning algorithm used to construct predictive features and update decision relevant descriptors from streaming data [84]. It can incorporate optimization enhanced machine learning to improve accuracy and robustness under operational constraints. In practical deployments, a learning module may be used to map raw or preprocessed sensing and early warning streams to intermediate quantities that support inference and decision making, such as short horizon demand proxies, control effectiveness indicators, or reliability scores for early warning features [85]. The quality of these learned quantities directly affects the informativeness of the exogenous inputs entering the latent structural system and can therefore influence posterior concentration and decision separability.

To strengthen this stage, hybrid optimization strategies can be adopted, coupling a fast learner such as an Extreme Learning Machine (ELM) with a metaheuristic optimizer, such as the grey wolf optimizer. In this configuration, the optimizer is used to tune key learning hyperparameters and representation choices, including hidden layer size, activation parameters, regularization strength, and feature selection, to minimize prediction error and improve generalization under noise and missingness [86]. The optimized learner can be trained offline using historical events and then deployed online to generate real time predictive descriptors, which are subsequently treated as observed inputs to the Bayesian model. This separation preserves probabilistic inference and uncertainty propagation in the core framework while improving the accuracy and stability of the upstream predictive layer.

From a decision making perspective, optimization enhanced learning is particularly relevant when warning lead time is short and rapid computation is required. Lightweight learners with optimized hyperparameters can provide low latency predictions that remain consistent with the feasibility constraints of the real time control loop. Future extensions may further integrate optimization objectives that penalize false alarms or late decisions to align the learning module with operational costs and code relevant decision thresholds, while retaining Bayesian posterior outputs for uncertainty aware control selection and compliance evaluation [87].

#### 2.7.5. Robustness of active and semi active control devices under component failures

Active and semi active control strategies are only beneficial if device level reliability supports timely and effective force delivery during the warning to response window. In practice, control performance can be degraded by partial component failures, including actuator saturation or loss of authority, valve or power supply malfunctions in semi active devices, sensor and controller faults that corrupt state estimation, and connection or anchorage degradation that reduces force transfer capacity. These failure modes motivate treating control system robustness explicitly as part of resilience assessment rather than as an external assumption [88].

Within the proposed framework, component failures are represented as operational degradation states that enter the model through observed diagnostics and through latent readiness mechanisms. Device health indicators, such as self-checks, command tracking errors, saturation events, and power availability, are incorporated as additional observed inputs and can also be mapped to the adaptive capability construct [38]. Similarly, reductions in force transfer efficiency due to connection distress are represented through sensing and communication integration and adaptive capability descriptors, reflecting that the effective control action depends on both command quality and reliable shear transfer at interfaces. As a result, failure conditions shift the posterior of the latent mechanisms toward lower readiness and increase posterior uncertainty, which propagates to lower expected compliance performance and to more conservative control mode selection when reliability is insufficient.

Robustness evaluation is implemented through scenario-based validation, in which the same warning and sensing conditions are simulated under nominal and degraded device states. Comparison targets include changes in posterior compliance probability, shifts in mode selection probabilities, and sensitivity of dominant structural pathways linking analytics quality and adaptive capability to the resilience outcome. This protocol directly addresses the practical question of whether the framework maintains consistent decisions and realistic performance predictions when control components are partially unavailable, and

clarifies that the proposed resilience gains are conditional on device reliability and connection level force transfer integrity rather than being assumed universally [89].

#### 2.7.6. Extension to multi hazard resilience, including fire following earthquake

Although the present demonstration focuses on earthquake induced demands, the proposed probabilistic architecture is directly extensible to multi hazard resilience, with material level damage characterization and liquefaction sensitive soil structure response considerations informing damage sensitive performance mapping within a unified inference structure [90,91]. A primary multi hazard case is compound post-event hazard escalation, where structural damage, utility loss, impaired egress, and degradation of protective system interact to elevate consequences beyond those implied by structural response alone, consistent with studies on the dynamic response and protective mechanisms of airbag-based mitigation systems [92].

Operationally, the extension is implemented by augmenting the observed input set with hazard coupling descriptors and expanding the outcome layer to include multi hazard performance targets. Additional inputs may include indicators of post event ignition likelihood, gas and electrical service disruptions, sprinkler and suppression availability, compartmentation integrity, and emergency access constraints [93]. These variables can be treated as observed covariates that influence early warning and post event operational reliability, and can also be incorporated as additional ordinal indicators that inform latent preparedness and functional continuity mechanisms [94]. The decision layer is expanded to permit multi objective actions, for example, simultaneous selection of a structural control mode and rapid safety actions such as shutdown of vulnerable utilities, activation of smoke control, and routing of occupants, subject to the same feasibility constraints governing computation and actuation within limited time windows [95].

Performance evaluation is generalized by defining a joint compliance objective that accounts for both structural and fire related limit states [19,96]. For example, the exceedance probability can be reformulated to represent the probability of violating either structural demand thresholds or post event fire safety thresholds, conditional on the inferred latent mechanisms and hazard coupling inputs. This formulation preserves end to end uncertainty propagation while enabling scenario based evaluation of how sensing quality, warning reliability, analytics readiness, and adaptive capability jointly influence resilience under coupled hazards [18,97]. Consequently, multi hazard decision making can be supported through posterior quantities for competing performance objectives and identification of readiness mechanisms providing the greatest leverage when earthquake consequences cascade into fire driven functional loss [98].

## 3. Results and discussion

### 3.1. Measurement model quality

The ordinal measurement model specified by Eqs. (8) to (11) yields consistently strong posterior target loadings across constructs, indicating stable identification of latent response characteristics under the threshold formulation [99]. As reported in Table 1, mean posterior loadings range from 0.70 to 0.80. The highest average loading is for SRP (0.80), followed by ACE (0.77), while the lowest average loading is for EWR (0.70). Reliability is high for all constructs. At the application level, however, resilience interpretation in ground-sensitive seismic systems should still recognize direction-dependent soil behavior rather than isotropic ground assumptions [100]. Composite reliability ranges from 0.83 for EWR to 0.92 for SRP, with relatively narrow credible intervals, supporting stable latent scaling under the identification constraint in Eq. (10) and the centered prior in Eq. (11). While these identification properties support measurement adequacy, SSI-sensitive seismic applications should still interpret response using constitutive soil formulations capable of representing nonlinear elastoplastic behavior [101]. Convergent validity is supported by AVE values that exceed 0.50 for every construct, with posterior means from 0.51 to 0.65, indicating that each construct explains a majority share of ordinal indicator variance on the latent response scale. Together, these results imply adequate threshold separation and informative discrimination among ordered categories, consistent with the requirement that the cut points $\tau_{m,r}$ remain strictly ordered in Eq. (8) [102].

In a broader structural sense, the need to distinguish among response patterns is also evident in vulnerable heritage systems subjected to dynamic impact [103]. Across all reported pairs, HTMT posterior means range from 0.53 to 0.63, and upper credible bounds do not exceed 0.72. The highest association is between RTAQ and ACE at 0.63 with a 95% credible interval from 0.54 to 0.72. The lowest association is between EWR and CSI, at 0.53, with a 95% credible interval of 0.43–0.63. These magnitudes support the empirical distinctiveness of the constructs under the ordinal specification. Treating the indicators as ordinal is structurally consistent with the threshold likelihood in Eq. (8) and (9) and avoids the implicit equal interval assumption that arises when Likert type responses are treated as continuous [104]. In this model, category probabilities are generated by the cut points $\tau_{m,r}$ rather than by linear distance on an observed scale, preserving the intended measurement level and reducing the risk of attenuated or inflated loadings when category spacing is nonuniform [105,106]. At the structural response level, reliable hysteretic behavior modeling of reinforced concrete walls under axial flexure shear interaction likewise remains important for resilience oriented performance assessment [107]. Table 4 reports a PPC tail area probability of 0.46 for the ordinal discrepancy, a polychoric residual RMSE of 0.07 for replicated versus observed associations, a maximum absolute item level PPC residual of 0.12, and a share of 0.14 of indicators with residuals exceeding 0.10. These checks indicate that the measurement layer reproduces the observed ordinal structure without systematic misfit that would typically emerge from continuous approximation of ordered outcomes. Table 1 shows strong measurement quality for all constructs. Average posterior loadings are high, from 0.70 to 0.80, composite reliability is strong, from 0.83 to 0.92 with tight 95 % credible intervals, and AVE indicates adequate convergent validity, from 0.51 to 0.65. SRP and ACE are the strongest overall, while EWR is comparatively lower but still acceptable.

Fig. 2 illustrates the measurement quality of the latent constructs RTAQ, EWR, ACE, CSI, and SRP using three complementary metrics. Average posterior loadings reflect the strength of indicator construct relationships, composite reliability with 95 % credible intervals indicates internal consistency, and average variance extracted with 95 % credible intervals captures convergent validity across constructs.

Table 2 reports posterior HTMT ratios for discriminant validity among RTAQ, EWR, ACE, and CSI. All pairwise HTMT means fall from 0.53 to 0.63, and all 95 % credible intervals remain below 0.72. This indicates satisfactory discriminant validity, with constructs empirically distinct and only moderate overlap, the strongest being between RTAQ and ACE, and between ACE and CSI.

Fig. 3 shows the posterior ordinal category probability profiles for EWR, ACE, and CSI across ordered response categories. The three constructs exhibit distinct probability patterns, indicating different ordinal response distributions across categories. EWR shows a sharp decline after the second category, indicating concentration of probability in

**Table 4**
PPCs and adequacy metrics.

| Diagnostic statistic | Posterior value |
|---|---|
| PPC tail area probability (ordinal discrepancy) | 0.46 |
| Polychoric residual RMSE (replicated vs. observed) | 0.07 |
| Max absolute item level PPC residual | 0.12 |
| Share of indicators with residual > 0.10 | 0.14 |

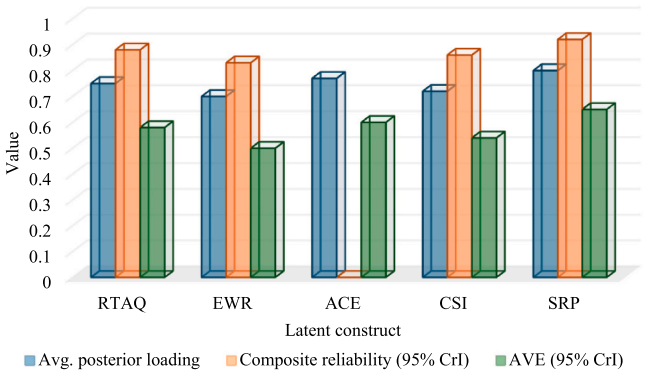



**Fig. 2.** Posterior loadings, composite reliability, and average variance extracted across latent resilience constructs.

lower ordinal states. ACE exhibits a gradual shift from moderate probability at lower categories to near zero probability at higher categories, reflecting transitional sensitivity. CSI shows a broader distribution, with a peak probability at an intermediate category and persistence into higher levels, indicating greater stability and gradation in construct expression. Table 3 indicates that all predictors have positive standardized effects on SRP with high posterior support. RTAQ has the strongest effect, with a posterior mean of 0.38, a 95% credible interval from 0.23 to 0.53, and Pr(beta> 0) = 0.998. ACE is next, with a mean of 0.31 and a credible interval from 0.18 to 0.45, and a probability of 0.999. At the same time, EWR and CSI are smaller but still positive, with means of 0.21 and 0.19, credible intervals excluding zero, and probabilities of 0.982 and 0.971. Posterior standard deviations are low, about 0.07–0.08, indicating stable estimates.

Fig. 4 presents the Bayesian regression results for RTAQ, EWR, ACE, and CSI as predictors of SRP. Posterior means indicate positive effect sizes for all predictors, with RTAQ and ACE contributing comparatively more to resilience performance. Posterior standard deviations are small, reflecting stable and well identified parameter estimates. The probability that each coefficient exceeds zero is close to one, providing strong posterior evidence of positive influence. Collectively, the results confirm that all four constructs significantly and robustly contribute to SRP.

### 3.2. *Structural path effects and key causal drivers*

Structural effects are interpreted under the latent system in Eq. (12), with standardized coefficients representing posterior means of directed relations among constructs and toward the outcome SRP. In the full model, all predictors show positive and credibly nonzero effects on SRP, with posterior probabilities $\Pr(\beta > 0)$ from 0.971 to 0.999 for all paths (Table 3). The largest direct driver is RTAQ with a posterior mean of 0.38 and a 95% credible interval from 0.23 to 0.53. The next largest effect is ACE at 0.31, with a 95% credible interval from 0.18 to 0.45. The remaining effects are EWR at 0.21 (95% credible interval: 0.07–0.36) and CSI at 0.19 (95% credible interval: 0.05–0.33; Table 3). This ordering indicates that RTAQ provides the primary leverage on SRP, followed by ACE, with EWR and CSI providing smaller but still substantively meaningful contributions. Model comparison underscores the importance of accounting for response style and allowing cross loadings to avoid distorted inferences. As shown in Table 5, the full BOSEM specification, which includes response style and cross loadings, achieves the smallest PSIS-LOOIC and WAIC values and receives the largest Bayesian stacking weight. Specifically, the full model attains PSIS-LOOIC 4121.6 and WAIC 4124.0, with stacking weight 0.75, outperforming the null structure and the restricted alternatives. This pattern indicates that the latent structure and its effects on SRP are best supported when measurement artifacts are explicitly modeled rather than absorbed into the structural relations in Eq. (12).

Endogeneity diagnostics further support causal interpretability within the stated boundary conditions. Table 6 reports posterior

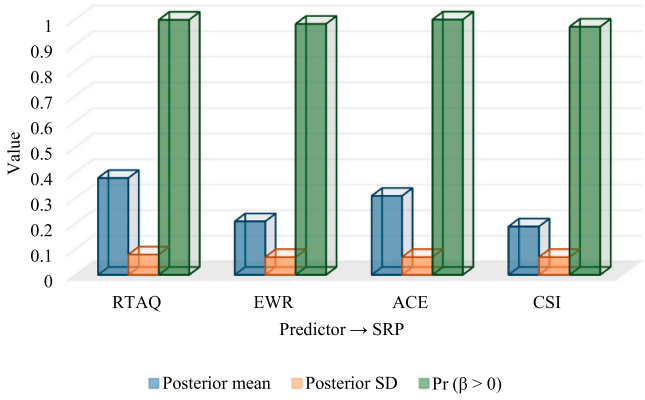



**Fig. 4.** Posterior coefficient summaries for the latent predictors of SRP, including posterior means, posterior standard deviations, and sign probabilities Pr (β > 0).

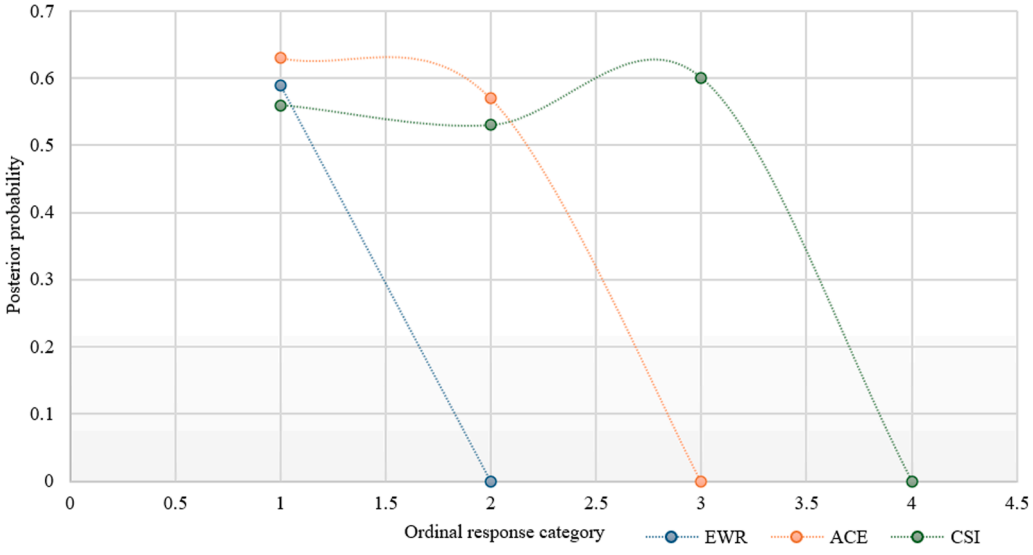



**Fig. 3.** Posterior probabilities across ordinal response categories for the latent constructs EWR, ACE, and CSI.

**Table 5**
Bayesian model comparison.

| Model specification | PSIS-LOOIC (SE) | WAIC (SE) | Bayesian stacking weight |
|---|---|---|---|
| Null structure (no RS, no CL) | 4215.3 (56.0) | 4218.7 (55.1) | 0.02 |
| Response style adjusted | 4163.9 (53.4) | 4166.5 (52.8) | 0.15 |
| Cross loading enabled | 4158.7 (52.9) | 4161.2 (52.1) | 0.08 |
| Full BOSEM (RS + CL) | 4121.6 (51.2) | 4124.0 (50.6) | 0.75 |

correlations between each predictor and the structural error, which are near zero, and all credible intervals span zero. The posterior means are 0.04 for RTAQ, 0.02 for EWR, minus 0.05 for ACE, and 0.06 for CSI. Instrument robustness metrics show nontrivial relevance for the instrumented components, with conditional $R^2$ equal to 0.16 for the instrument predicting RTAQ and 0.12 for the instrument predicting CSI, with credible intervals indicating meaningful but not deterministic explanatory power (Table 6). Collectively, these results support a mechanism in which gains in SRP are most strongly associated with improvements in RTAQ and ACE. At the same time, EWR and CSI contribute additional incremental effects, and the overall model is strengthened when response style and cross loading structure are included. Table 4 indicates good model adequacy, with a near uniform PPC tail area probability of 0.46, a low polychoric residual RMSE of 0.07, a limited maximum item level residual of 0.12, and only 0.14 of indicators exceeding 0.10 in residual. Table 5 shows clear predictive fit improvement across specifications, with the full BOSEM including response style and cross loading achieving the lowest PSIS-LOOIC and WAIC, 4121.6 and 4124.0, and the dominant stacking weight of 0.75. Table 6 suggests limited endogeneity, as correlations between each latent predictor and its structural error are close to zero and all 95 % credible intervals include zero. Instrumental strength is moderate, with a conditional R squared of 0.16 for RTAQ and 0.12 for CSI, supported by credible intervals indicating nontrivial explanatory power.

Fig. 5 compares model specifications using PSIS-LOOIC and WAIC, with standard errors as measures of predictive performance. The null structure shows the poorest fit, while the response style adjusted model improves WAIC but not PSIS-LOOIC. Allowing cross loading yields further improvement, indicating better representation of measurement complexity. The full BOSEM specification, with response style and cross loading, achieves the lowest information criteria values, demonstrating superior out of sample predictive adequacy.

Fig. 6 presents correlations between structural errors and the conditional R-squared of the instrumental equation. Correlations for RTAQ and EWR structural errors are small and positive, while ACE shows a slight negative correlation, and CSI exhibits a modest positive association. These low magnitude correlations suggest limited endogeneity and acceptable model specification. The conditional R-squared values indicate moderate explanatory power for the instruments predicting RTAQ and CSI, with stronger explanatory power for RTAQ.

**Table 6**
Endogeneity assessment and instrumental robustness.

| Quantity | Posterior mean | 95% CrI |
|---|---|---|
| Cor (RTAQ, structural error) | 0.04 | [-0.11, 0.18] |
| Cor (EWR, structural error) | 0.02 | [-0.13, 0.16] |
| Cor (ACE, structural error) | -0.05 | [-0.20, 0.09] |
| Cor (CSI, structural error) | 0.06 | [-0.09, 0.21] |
| Conditional $R^2$ of instrument → RTAQ | 0.16 | [0.08, 0.26] |
| Conditional $R^2$ of instrument → CSI | 0.12 | [0.05, 0.22] |

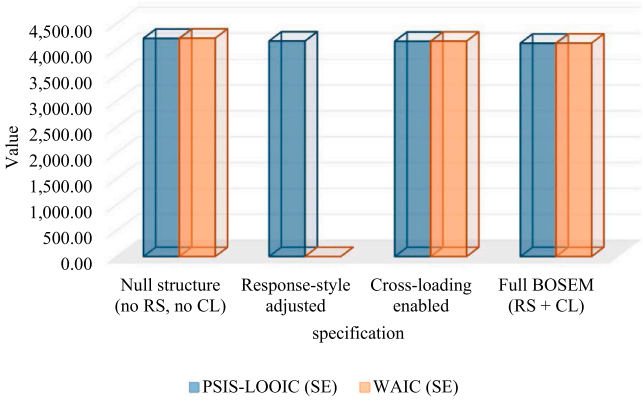



**Fig. 5.** Model comparison across Bayesian ordinal specifications using PSIS LOOIC and WAIC uncertainty estimates.

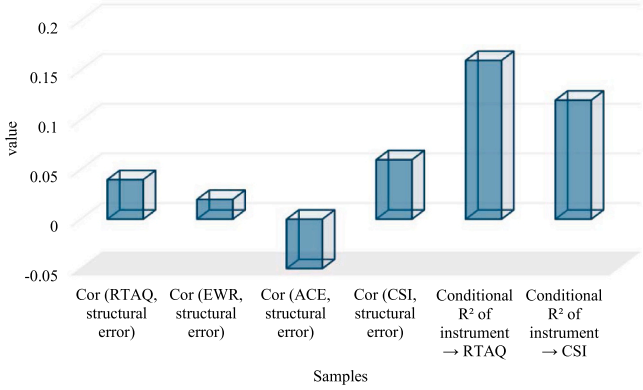



**Fig. 6.** Structural error correlations and conditional R-squared diagnostics for the resilience structural model and instrumental robustness assessment.

### 3.3. Uncertainty propagation from early warning and sensing

In this study, the early warning reliability channel is represented by EWR, the sensing and communication integration channel by CSI, the real time analytics quality channel by RTAQ, adaptive capability by ACE, and the resilience outcome by SRP (Tables 1–3). Uncertainty enters the seismic demand channel through the intensity error model in Eq. (13) and propagates to compliance through Eq. (15). As $\sigma_I$ increases, the posterior for $I^{\text{true}}$ broadens, increasing posterior dispersion in $r = \Pr(D > D_{\text{lim}} \mid \eta, I^{\text{true}}, c)$, even when the posterior for $\eta$ is stable. This mechanism is probabilistic rather than structural, with $\sigma_I$ perturbing the demand driver while the ordinal measurement scale defined by Eqs. (8) to (11) is left unchanged. Under this structure, higher $\sigma_I$ increases outcome uncertainty and reduces decision certainty in borderline cases where $D$ is near $D_{\text{lim}}$, as posterior mass shifts more frequently across the limit state in Eq. (15).

Sensing and communication degradation enter through the boundary constrained variables in Eq. (2) and the input construction in Eq. (5), and then affect the latent system through Eq. (12). Within the construct set reported in Tables 1–3, CSI represents this channel. Its structural association with SRP is summarized in Table 3. The expected direction of propagation is an implication of the model equations: larger latency and higher packet loss reduce effective information content, which weakens the information entering the latent system in Eq. (12), and this in turn reduces decision certainty in Eq. (14) and increases uncertainty in the compliance probability in Eq. (15). This statement is a theoretical implication of the specified channels in Eqs. (2), (5), (12), (14), and (15), rather than a separate empirical estimate reported in Tables 1–6.

Boundary condition testing in Section 2.7.2 supports using the full

specification as the baseline for propagation studies. Table 5 shows that the full BOSEM specification, including response style and cross loadings, provides the best expected predictive fit, with a PSIS-LOOIC of 4121.6 and a Bayesian stacking weight of 0.75. These findings support evaluating uncertainty propagation under the same measurement layer used for inference, rather than under simplified alternatives that may confound measurement artifacts with structural uncertainty. Table 6 supports the conclusion that the residual dependence between each predictor and the structural disturbance is weak, since the posterior correlations between RTAQ, EWR, ACE, and CSI and the structural error are near zero, and their credible intervals span zero. This limits the risk that apparent propagation patterns reflect unmodeled confounding rather than the explicit pathways in Eqs. (12) and (13). Patterns across building typology and occupancy classes arise through the contextual vector $c$ in Eq. (5) and its direct role in compliance in Eq. (15). This context can also include site dependent geotechnical conditions in ground sensitive applications, where large strain soil behavior and CPTU/SBTn-based characterization may influence how uncertainty is interpreted in resilience assessment [108]. Higher importance classes and more stringent performance objectives effectively reduce tolerance for exceedance, so the same perturbation in $I^{true}$ or in $\eta$ produces a larger practical change in posterior risk near decision thresholds. This mechanism is implied by Eq. (15) through the conditioning on $c$ and does not require changes in the ordinal measurement layer.

### 3.4. Control mode behavior and decision consistency

Control mode selection is governed by Eq. (14), where $\Pr(d_k = d|\eta, w_k)$ depends on the latent mechanisms and current warning features. Decision consistency follows from two conditions supported by the reported measurement results. First, the constructs are measured with high reliability and adequate discriminant validity, as shown in Eqs. (8) to (11), as shown by the composite reliability and AVE results in Table 1 and the HTMT results in Table 2. Second, posterior predictive adequacy for the ordinal indicators is acceptable, as reported in Table 4, thereby limiting the risk that variability in mode selection is driven by measurement misfit rather than by uncertainty transmitted through the warning and sensing channels.

The structural effect magnitudes in Table 3 provide the empirical basis for identifying which latent mechanisms have the strongest direct associations with SRP within the specified structural system in Eq. (12). RTAQ has the largest standardized direct effect on SRP, followed by ACE, with EWR and CSI also showing positive effects. Accordingly, within the decision layer in Eq. (14), mode selection that assigns a higher posterior probability to active strategies is expected to be most consistent when the posterior for the key mechanisms with larger structural effects on SRP is high. Conversely, increases in uncertainty in the warning features $w_k$ and reductions in the effective information entering the latent system through the sensing channel are expected, by the structure of Eq. (14), to reduce separation among posterior mode probabilities and shift posterior mass toward more conservative modes. These directions of effect statements are model based implications of Eqs. (12) and (14), not additional empirical quantities reported in Tables 1–6.

Comparisons among passive, semi active, and adaptive passive modes follow directly from the feasibility constraints in Eqs. (3) and (4). When latency $L_k$ increases, the constraint in Eq. (4) becomes more restrictive, reducing the feasible region for timely adaptive actions even if the posterior for $\eta$ is favorable. In that regime, the decision layer in Eq. (14) remains probabilistic but is operationally filtered toward modes that require less actuation time or less reliance on rapid updates. When sensing quality degrades due to missingness and packet loss, the effective information entering the latent system through the channel represented by CSI is reduced, thereby diminishing the posterior advantage of modes that rely on precise and timely state estimation. Under improved warning information and stronger sensing and communication integration, separation among mode probabilities increases, as both $\eta$ and $w_k$ provide more consistent evidence in Eq. (14), supporting repeatable selection of semi-active or adaptive passive strategies when feasibility under Eq. (4) is maintained.

### 3.5. Code relevant performance and compliance probability

Code relevant compliance is quantified by the exceedance probability $r = \Pr(D > D_{\lim}|\eta, I^{\text{true}}, c)$ in Eq. (15), with $I^{\text{true}}$ generated by the intensity uncertainty model in Eq. (13) and the latent drivers $\eta$ governed by the structural system in Eq. (12). Across scenarios, the dominant determinant of posterior shifts in $r$ is the joint movement of the posterior distribution of $\eta$ and the dispersion of $I^{\text{true}}$. Scenarios with higher early warning uncertainty $\sigma_I$ broaden the posterior of $r$ and increase tail mass above the compliance threshold, as the distribution of $D$ overlaps $D_{\lim}$ more frequently, particularly near the limit state where small intensity perturbations translate into exceedance events. Scenarios with degraded sensing and communication, represented by higher latency $L_k$ higher packet loss, and missingness embedded in $s_k$ with latency and packet loss constrained by Eq. (2), shift the posterior of $\eta$ through the $\Gamma x$ term in Eq. (12), reducing effective information quality and compressing posterior support for real time analytics and control readiness. This mechanism increases $r$ through increased uncertainty in decision relevant constructs and weakened downstream resilience pathways.

Mode dependence in compliance follows from the decision layer in Eq. (14) coupled with the feasibility limits in Eqs. (3) and (4). When the posterior assigns greater probability to semi active or adaptive passive modes, the implied control actions reduce the conditional exceedance probability $r$ relative to passive operation, if actuation remains feasible within the warning window. The expected improvement is largest when the posterior for real time analytics and control readiness is high, consistent with the dominant standardized effects reported in Table 3 for RTAQ $\beta = 0.38$and ACE $\beta = 0.31$ In contrast, when $L_k$ approaches the operational limit implied by Eq. (4), the relative advantage of adaptive modes diminishes, as timely execution becomes less probable and the posterior distribution of feasible control actions shifts toward more conservative levels, thereby increasing $r$ toward passive like levels even when latent readiness is favorable.

The ordinal performance state $z$ in Eq. (7) is obtained by mapping the continuous risk output $r$ into objective aligned categories, as described following Eq. (15). Scenario and mode effects, therefore, appear as distributional shifts in $z$ that reflect movement of posterior mass across the probability thresholds defining low, moderate, and high-performance states. Improvements in data acquisition and transmission quality and reductions in early warning uncertainty shift the posterior distribution toward lower exceedance probability, increasing the posterior probability of the Low state and reducing the High state mass. Degradation in sensing quality, increased latency, or increased $\sigma_I$ produces the opposite shift, with a larger fraction of posterior draws falling into moderate and high categories. This ordinal distribution shift is not an additional modeling assumption beyond Eq. (15) but a direct consequence of posterior risk movement under the uncertainty propagation channels defined by Eqs. (12) and (13) and the decision dependence in Eq. (14).

Quantitative reporting of scenario level trends in $r$ and in the posterior state probabilities $\Pr(z = \text{Low})$, $\Pr(z = \text{Moderate})$, and $\Pr(z = \text{High})$ requires the scenario outputs by mode. The posterior means and credible intervals of $r$ by scenario and by control mode, together with the frequency of each $z$ category in posterior predictive draws, can be reported as explicit numeric contrasts with table references in the same style as Tables 3–6. Within the BOSEM framework integrates IoT sensing quality, early warning reliability, real time analytics capability, adaptive control readiness, and code and standards integration within a unified probabilistic structure. The implemented methodology combined an ordinal threshold measurement layer for survey and inspection indicators in Eqs. (8) to (11), with a latent structural system linking

constructs and aggregated sensing and warning features in Eq. (12), an explicit early warning intensity uncertainty model in Eq. (13), and a probabilistic decision and compliance layer in Eqs. (14) and (15). The system boundary was defined over the early warning to post action window, with latency and packet loss constraints in Eq. (2) and actuation feasibility limits in Eq. (4), thereby ensuring that inference and control remained operationally admissible. Bayesian inference was performed over the full parameter set with PPCs and model comparison to validate measurement adequacy, structural consistency, and robustness to uncertainty.

Results indicated strong ordinal measurement performance, with high composite reliability and adequate convergent and discriminant validity across constructs, and posterior predictive diagnostics indicating good reproduction of category probabilities. In the structural layer, real time analytics capability and adaptive control readiness emerged as the dominant drivers of the resilience outcome, with the largest standardized effects on the compliance related construct. Early warning reliability and code integration provided additional, though smaller, direct contributions. Model comparison favored the full specification with response style and cross loading adjustments, and endogeneity diagnostics indicated limited residual confounding. Uncertainty propagation analyses showed that increases in early warning intensity variance, latency, and data degradation expanded posterior uncertainty in exceedance probability and shifted performance state distributions toward less favorable categories, particularly under constrained actuation windows. Control mode selection responded coherently to changes in latent constructs and warning quality, with adaptive strategies providing the greatest reduction in exceedance probability when feasibility conditions were satisfied. Overall within the evaluated scenarios, resilience improvement was maximized when high quality sensing and warning information were effectively transformed into reliable analytics and executable control actions within operational time constraints. To place these findings in context, the following studies illustrate how related approaches address selected parts of the sensing to warning to control chain, while remaining distinct from the end to end probabilistic structure of BOSEM.

Recent studies typically advance only one segment of the sensing to warning to control chain, but rarely provide a single posterior that connects information quality to code relevant compliance, as in Eqs. (12) to (15). Milovančević et al. [109] present an object oriented e-diagnostics information system for dynamical monitoring, using UML-based diagrams to represent sensing, vibration analysis, and monitoring system behavior. Their data are measured acceleration and vibration information together with system architecture and behavioral representation, and performance is evaluated through monitoring and diagnostic capability rather than inference of latent readiness mechanisms. Uncertainty is treated implicitly through observed performance variability rather than explicitly via a hazard error layer such as Eq. (13) and a compliance probability definition, such as Eq. (15). The gap relative to BOSEM is the lack of an ordinal measurement layer for constructs and an end to end posterior mapping from sensing and communication quality to decision behavior and compliance .

Abdalzaher et al. [110] review machine learning and IoT directions for EEW in smart cities and consolidate pipeline components and evaluation issues across prior studies. The data evidence is comparative, and literature based, and the contribution is architectural and algorithmic synthesis rather than a causal latent variable model. Uncertainty is discussed as a performance consideration but not modeled for propagation into downstream operational decisions. The gap relative to BOSEM is the lack of ordinal latent constructs using Eqs. (8) to (11) and the lack of a compliance target formulated as posterior exceedance probability in Eq. (15).

Marcou et al. [111] analyze end to end EEW performance in large earthquake cases using dense, distributed sensing and fast delivery concepts to evaluate alerting against shaking arrival. Their data are EEW system outputs and time dependent performance measures, and the focus is on system level effectiveness rather than on building specific decision and control. Uncertainty is expressed through the variability of EEW performance in rare large events, but it is not formalized as a probabilistic intensity error model of the form in Eq. (13) coupled to a code relevant risk metric as in Eq. (15). The gap relative to BOSEM is the missing linkage between information quality, latent operational mechanisms, and probabilistic compliance outcomes under explicit feasibility constraints such as Eq. (4).

A second benchmark stream expands sensing coverage and improves warning reach, but still falls short of control consistency and PBEE aligned performance mapping. Similarly, Allen et al. [112] report the methodology and performance of Android based global earthquake detection and alerting using smartphones, with evaluation over multiple years and large scale deployments. The data are crowdsourced accelerometer signals and alert outcomes, and the contribution is scalable detection and dissemination rather than building level control. Uncertainty is managed through algorithm design and empirical performance evaluation rather than through an explicit posterior hazard and decision coupling that conditions mode selection on latent readiness. The gap relative to BOSEM is that warning generation is not integrated with an ordinal latent mechanism model and does not yield a posterior compliance probability defined as Eq. (15) conditioned on operational constructs and building context.

Tiggeloven et al. [73] review the role of AI in early warning systems within a four pillar framework that includes preparedness for response. The study is literature based and focuses on applicability and guardrails rather than a building level probabilistic control model. Uncertainty is emphasized as a requirement for actionable early warnings. Still, the review does not instantiate an uncertainty propagation chain from sensing and warning to operational decisions, subject to constraints such as Eqs. (2) and (4). The gap relative to BOSEM is the absence of a structural pathway system, as in Eq. (12) that quantifies mechanism effects and links them to decision probabilities and compliance probability.

A third benchmark stream targets vibration mitigation and semi active devices, often demonstrating response reduction but rarely embedding real time warning uncertainty and feasibility limits in a unified probabilistic decision layer. In vibration control studies, Katebi et al. [113] present a comparative analysis of metaheuristic optimization algorithms for optimal active control of structures, in which six intelligent optimization methods are coupled with a wavelet based LQR scheme to improve feedback gains under earthquake excitation. The evidence base is computational and comparative, focusing on structural response reduction, convergence rate, and computational effort rather than on a posterior decision rule conditioned on warning features and latent mechanisms. Uncertainty is typically treated through scenario comparisons rather than through explicit Bayesian propagation from hazard error and information quality into decisions and compliance. The gap relative to BOSEM is the missing connection between readiness constructs, warning reliability, and probabilistic mode selection in Eq. (14) under feasibility conditions like Eq. (4).

Daie et al. [114] investigate the use of pre bent steel strips as seismic dampers through finite element simulations of braced and unbraced frame systems under dynamic loading and benchmark earthquakes. Their data are numerical structural response outputs, including base shear, story acceleration, story velocity, and story displacement, used to demonstrate the effectiveness of the damping device rather than a controller level decision formulation. Uncertainty is not propagated from warning error or network degradation into decision and performance states. The gap relative to BOSEM is the absence of the joint decision and compliance mapping in Eqs. (14) and (15) conditioned on latent constructs inferred from ordinal indicators.

Qin et al. [115] analyze the progressive collapse resistance of steel beam-column structures with column tree connections under edge-column-failure scenarios. The data are structural models and collapse-resistance simulations used for robustness style evaluation,

which aligns with the motivation for probabilistic treatment of performance. However, the study does not incorporate an EEW information layer or a latent construct layer for readiness, nor does it incorporate a probabilistic mode selection rule conditioned on real time warning features. The gap relative to BOSEM is the absence of end to end uncertainty propagation from sensing and warning through constructs into operational mode decisions and code relevant compliance probability, as in Eq. (15).

A fourth benchmark stream addresses probabilistic decision support and ordinal outcome modeling, which partially overlaps with BOSEM components but does not integrate the full sensing to decision to compliance chain. At the decision support level, Yazdani et al. [116] propose a simheuristic approach for improving construction and demolition waste collection service under uncertainty, using an integrated simulation-optimization framework in which a hybrid Genetic Algorithm is applied to optimize vehicle route planning between construction projects and recycling facilities. The data consist of operational routing, facility allocation, travel time, and capacity related information, and uncertainty is handled within a simulation-based optimization framework rather than through latent seismic readiness mechanisms. Still, the decision scope is at the planning and service optimization level rather than operational control, subject to latency and actuation constraints. The study does not incorporate a real time sensing quality layer or a control mode selection model of the form in Eq. (14), and it does not target code relevant exceedance probability conditioned on operational constructs. The gap relative to BOSEM is the operational scale coupling of Eqs. (2) to (4), Eq. (12), and Eq. (15) within a single posterior . Collectively, these studies provide partial benchmarks for sensing, warning, control, and decision support, but none offers the integrated ordinal probabilistic compliance mapping developed in the present BOSEM framework.

Chen develops generalized ordinal regression models for seismic fragility analysis as an alternative to conventional fragility fitting, using ordinal damage state data and model comparison. The method directly supports ordinal outcome modeling logic consistent with the threshold mapping in Eq. (8) and avoids continuous treatment of ordered categories. Uncertainty is treated statistically within the fragility model, but the scope remains focused on fragility estimation rather than on causal pathways among readiness mechanisms and operational decisions. The gap relative to BOSEM is that ordinal modeling is not embedded within a latent construct system in Eq. (12) and is not coupled to mode selection and compliance probability in Eqs. (14) and (15) under sensing and warning constraints [117].

The present contribution is the development of an operationally constrained BOSEM framework that unifies measurement validity, causal mechanism inference, and code relevant performance evaluation within a single posterior. Unlike typical IoT and EEW studies that focus on detection latency or alert accuracy, the proposed formulation embeds sensing and communication limits through Eqs. (2) to (4), represents survey and inspection indicators using an ordinal threshold likelihood in Eqs. (8) to (11), and links these constructs to downstream operational mechanisms through the structural system in Eq. (12). Early warning intensity is treated as uncertain through Eq. (13), enabling end to end uncertainty propagation from warning and sensing impairments to decisions and performance. This structure yields decision grade outputs via the probabilistic mode selection model in Eq. (14) and the compliance metric $r$ in Eq. (15), which aligns resilience evaluation with performance based earthquake engineering rather than qualitative readiness scores.

Empirically, the results demonstrate that the ordinal measurement layer is reliable and discriminant, supporting construct interpretability and avoiding bias introduced by treating ordered indicators as continuous. The structural results identify real time analytics capability and adaptive control readiness as the dominant causal drivers of the resilience outcome, establishing a mechanism based prioritization logic for deployment and retrofit investment. Model comparison and posterior predictive diagnostics support the need to account for response style and cross loadings to prevent measurement artifacts from contaminating causal pathways. The combined evidence supports a coherent interpretation that resilience gains depend less on the mere presence of warning information and more on the ability to convert warning and sensing streams into reliable inference and executable control within the feasible time margin imposed by latency and lead time. In practical terms, the work provides a quantified basis for minimum operational feasibility requirements and for ranking interventions that improve sensing and communication quality, analytics capability, and control readiness to achieve code aligned performance objectives under uncertainty.

### 3.6. Economic feasibility and false alarm costs

Economic feasibility of IoT enabled resilience depends on limiting unnecessary actions triggered by uncertain information, as false alarms can impose nontrivial costs through avoidable evacuations, operational interruptions, service disruptions, and premature activation of control devices. Accordingly, decision rules must balance risk reduction against operational cost of acting when damaging shaking does not materialize or when inferred severity is overstated. The proposed framework supports cost aware operation, as posterior decision quantities are produced rather than only point estimates. In practice, actions can be triggered through probability thresholds on the compliance related exceedance metric, so that control activation or operational shutdown is initiated only when posterior risk exceeds a prespecified level. These thresholds can be selected using an expected cost criterion penalizing false positives and false negatives differently, reflecting that cost of an unnecessary action may be lower or higher than cost of a delayed action, depending on occupancy class and functional criticality. The same logic extends to control mode selection through decision preferences accounting for actuation cost and disruption, thereby discouraging aggressive modes when posterior evidence is weak. As sensing degradation and warning uncertainty increase posterior dispersion, conservative operation under limited credibility information is also enabled without systematic overreaction being forced. When data quality indicators and early warning uncertainty imply low decision certainty, posterior probabilities for high consequence actions typically become less concentrated, supporting delayed or limited interventions unless risk remains credibly high. This design aligns resilience improvement with practical cost constraints by reducing avoidable actions while preserving timely response when posterior evidence supports elevated hazard and compliance risk.

## 4. Conclusion

This study introduced a BOSEM framework that integrates IoT sensing and communication quality, earthquake early warning reliability, real-time analytics quality, and adaptive control readiness into a unified probabilistic assessment of code-relevant seismic resilience. By jointly modeling ordinal survey and inspection indicators with latent system mechanisms and operational constraints, the framework links data quality and decision readiness directly to resilience outcomes under uncertainty.

Empirical results from a sample of 162 observations demonstrate strong measurement performance and construct distinctness. Mean posterior loadings ranged from 0.70 to 0.80, composite reliability values ranged from 0.83 to 0.92, and average variance extracted ranged from 0.51 to 0.65 across constructs. Discriminant validity was supported by posterior HTMT means between 0.53 and 0.63, with upper credible limits below 0.72. Posterior predictive checks indicate adequate model fit, with a tail-area probability of 0.46, a polychoric residual RMSE of 0.07, a maximum item-level residual of 0.12, and only 14% of indicators exceeding a residual magnitude of 0.10.

Structural results indicate that real-time analytics quality and adaptive control readiness are the strongest direct drivers of seismic

resilience performance. Standardized effects on the resilience outcome were 0.38 for real-time analytics quality and 0.31 for adaptive control readiness, compared with 0.21 for early warning reliability and 0.19 for sensing and communication integration. All effects were positive, with posterior probabilities exceeding 0.97, indicating strong and consistent evidence of influence. Model comparison metrics further favored the full BOSEM specification, which achieved lower predictive information criteria and a stacking weight of 0.75 relative to alternative model formulations.

Diagnostic analyses indicate limited residual dependence and weak endogeneity, with posterior mean residual correlations close to zero and credible intervals spanning zero for all predictors. Instrument relevance was nontrivial, with conditional $R^2$ values of 0.16 for real-time analytics quality and 0.12 for sensing and communication integration, supporting the robustness of the structural inferences.

Overall, the results indicate that resilience improvement is primarily driven by strengthening real-time analytics capability and adaptive control readiness, while sensing integration and early warning reliability provide meaningful secondary contributions. The proposed framework offers a decision grade basis for prioritizing interventions by quantifying how sensing quality, early warning uncertainty, and operational constraints propagate to resilience performance. This approach provides a probabilistic framework for data-informed decision support for smart and resilient infrastructure systems.

## CRediT authorship contribution statement

**Zhonghong Li**: Writing – review & editing, Formal analysis, Investigation, Methodology, Software, Validation. **Gongxing Yan**: Investigation, Methodology, Software, Validation, Writing – review & editing. **Yousef Zandi**: Software, Resources, Methodology. **Morteza Gholizadeh**: Writing – original draft, Methodology, Investigation, Funding acquisition. **Yang Chen**: Writing – original draft, Visualization, Data curation, Conceptualization. **Yasmin Khairy**: Visualization, Validation, Supervision, Software. **José Escorcia-Gutierrez**: Software, Resources, Methodology, Investigation.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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